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・第 n 次導関数

(例) 次の関数の第2次導関数, 第3次導関数を求めよ。

(1) $y = x^3 - 2x^2 + x + 1$

$$y' = 3x^2 - 4x + 1$$

$$y'' = 6x - 4 \quad ,$$

$$y''' = 6 \quad ,$$

(2) $y = \frac{1}{2x-1}$

$$y = (2x-1)^{-1} \text{ として}$$

$$y' = -(2x-1)^{-2} \cdot 2 = -2(2x-1)^{-2}$$

$$y'' = -2 \cdot (-2)(2x-1)^{-3} \cdot 2 = 8(2x-1)^{-3} = \frac{8}{(2x-1)^3} \quad ,$$

$$y''' = 8 \cdot (-3)(2x-1)^{-4} \cdot 2 = -48(2x-1)^{-4} = -\frac{48}{(2x-1)^4} \quad ,$$

(3) $y = \sin 3x$

$$y' = \cos 3x \cdot 3 = 3 \cos 3x$$

$$y'' = 3 \cdot (-\sin 3x) \cdot 3 = -9 \sin 3x \quad ,$$

$$y''' = -9 (\cos 3x) \cdot 3 = -27 \cos 3x \quad ,$$

(4) $y = x e^x$

$$y' = 1 \cdot e^x + x e^x = (x+1) e^x$$

$$y'' = 1 \cdot e^x + (x+1) e^x = (x+2) e^x \quad ,$$

$$y''' = 1 \cdot e^x + (x+2) e^x = (x+3) e^x \quad ,$$